

AI-Powered Tea Quality and Moisture Prediction with Intelligent Auction Price Forecasting

Project ID: 25-26J-292

Project Proposal Report

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Bachelor of Science Special Hons Degree in Information Technology

Sri Lanka Institute of Information Technology

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Automated Tea Leaf Inspection Using Computer Vision and Machine Learning

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor: Dr. Bhagya Nathali Silva	Date: 29.08.2025	
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Abstract

Tea quality assessment plays a vital role in maintaining product value, market competitiveness, and consumer satisfaction in the global tea industry. As part of the broader research project aimed at automating quality control in tea production, this component focuses on developing an advanced, real-time system for detecting and classifying tea plucking stages — under-plucked, correctly plucked, and over-plucked — through image processing and machine learning. Traditional inspection methods rely heavily on manual observation, which is prone to subjectivity, inconsistency, and inefficiency, ultimately affecting overall quality and productivity. The proposed solution introduces a conveyor-based image capture setup designed to acquire high-quality, standardized images of fresh tea leaves under controlled lighting and positioning conditions. Captured images will be labeled to create a robust dataset for supervised learning, enabling the training of advanced machine learning models to classify plucking stages using visual cues such as bud-to-leaf ratio, leaf size, color, and edge sharpness. To strengthen practical applicability, a worker identification module will be integrated, linking classification results with individual pluckers for automated performance tracking and efficiency analysis. This approach not only ensures objective, real-time assessment of plucking quality but also generates actionable insights for workforce optimization, targeted training, and continuous improvement. By replacing traditional inspection methods with data-driven, scalable technology, the component directly contributes to the overarching project goal of enabling predictive analytics in estate management. The anticipated outcome is a practical and deployable system that enhances tea quality consistency, improves labor productivity, and supports informed decision-making, ultimately driving sustainable operational growth in the tea industry.

Keywords: Tea plucking classification, image processing, machine learning, conveyor-based system, worker performance tracking, estate management, predictive analytics.

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Introduction

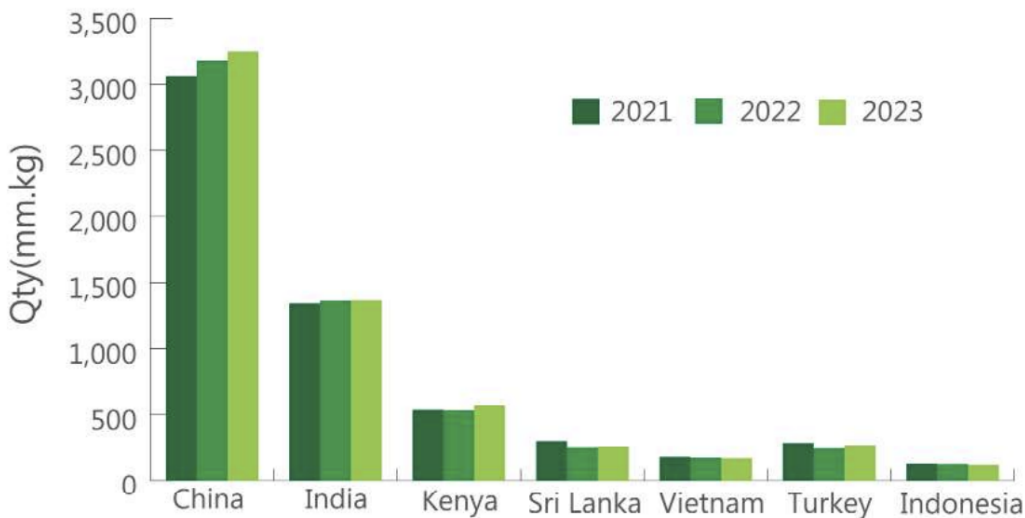
1.1 Background

Tea cultivation holds a central place in Sri Lanka's economy, culture, and identity. Since its introduction by James Taylor in the mid-19th century, Sri Lanka—then Ceylon—has become one of the world's leading producers and exporters of tea, with Ceylon Tea being recognized globally for its unique quality and flavor [1]. The industry contributes significantly to foreign exchange earnings, rural employment, and sustainable livelihoods across multiple regions, especially in the central highlands.

Over the years, extensive research has been conducted on improving tea cultivation, addressing productivity challenges, and ensuring long-term sustainability. Studies have examined aspects such as soil fertility management, pest and disease control, climate change impacts, and the adoption of mechanization and digital technologies in plantations [2], [3]. Sri Lankan researchers have also focused on enhancing tea quality through biochemical studies, value-added tea products, and process innovations in plucking and withering [4].

At present, the tea industry faces new challenges, including fluctuating global prices, rising production costs, labor shortages, and climate variability. To address these, the state of the art involves integrating modern technologies such as precision agriculture, IoT-based monitoring systems, machine learning for yield prediction, and sustainable farming techniques [5]. While global research on smart farming has shown promising results, its application within Sri Lankan tea plantations remains limited and requires adaptation to local conditions.

: World Tea Production Statistics



Source: Supplement: Annual ITC Bulletin of Statistics - 2023

Figure 1 - World Tea Production Statistics

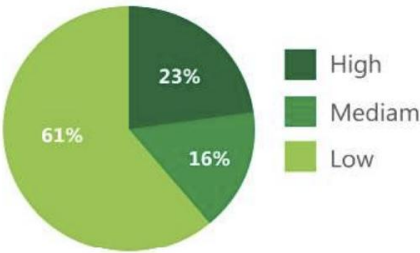
Tea is the world's second most consumed beverage, with global production reaching approximately 6,422.66 million kg in 2022, of which just over 48% originated from China, followed by India with about 21%, and other major contributors including Kenya and Sri Lanka [7]. Together, China, India, Kenya, and Sri Lanka account for an impressive 81% of total world tea production [8]. Beyond production, the tea export market is dominated by a similar group: China, Sri Lanka, Kenya, and India, which together generate over 70% of global tea export value, with China alone valued in billions of dollars annually [9],

[10]. This global landscape highlights Sri Lanka’s continued relevance, yet also underscores the immense pressure to maintain quality and competitiveness amid industry giants.

Elevation	Qty kg
High	58,639,860
Medium	42,337,896
Low	155,110,890
Total	256,088,647

Source: Sri Lanka Tea Board

Chart 3.1: Total Tea Production by Elevation



Source:Sri Lanka Tea Board

Figure 2 - Total Tea Production By Elevation

Sri Lanka’s tea production also varies considerably by elevation, which directly influences flavor, quality, and market value. As shown in Figure 1, low-grown regions contribute the highest share to total production (155,110,890 kg), followed by high-grown regions (58,639,860 kg) and medium-grown regions (42,337,896 kg), leading to a combined total of 256,088,647 kg annually. This distribution highlights both the economic dominance of low-grown tea and the importance of protecting highland cultivation for premium-quality exports.

Previous attempts to solve these challenges have included government-led initiatives, research by the Tea Research Institute of Sri Lanka (TRI), and collaborations with universities to introduce sustainable practices [6]. However, gaps remain in applying advanced data-driven approaches, such as predictive modeling and decision-support systems, to optimize plantation management. This research builds on earlier work by proposing an innovative framework that combines modern computational techniques with domain-specific knowledge of Sri Lankan tea cultivation. By doing so, it seeks to bridge the gap between traditional practices and emerging technological solutions.

1.2 Literature Survey

1.2.1 Climate and Soil Management

Climate variability has a direct impact on tea yield, biochemical composition, and overall quality. Herath and Abeysinghe [10] studied temperature and rainfall patterns across Sri Lankan tea regions, showing significant correlations between fluctuating microclimates and catechin levels, which directly influence tea flavor and market value. Soil fertility management is equally critical, with research emphasizing the importance of organic amendments, precise nutrient balancing, erosion control, and sustainable soil conservation practices [11]. Despite these findings, current plantation management often relies on manual observation and empirical experience, leading to suboptimal resource allocation and inconsistent quality. This underscores the need for data-driven soil and climate monitoring systems that provide real-time insights for decision-making, reducing losses due to climate-induced variability.

1.2.2 Mechanization and Digital Technologies

Mechanization in tea cultivation has evolved to include automated plucking, sorting, and processing technologies aimed at improving operational efficiency without compromising leaf quality [12]. Recent studies propose IoT-enabled monitoring systems that collect real-time data on soil moisture, temperature, pest incidence, and leaf condition [13]. These digital technologies allow for precision irrigation, fertilization, and pest control, reducing resource wastage and improving sustainability. However, adoption within Sri Lankan plantations remains limited, partly due to cost constraints and a lack of integrated frameworks linking sensor data with actionable decision-making tools. Research indicates that combining IoT with machine learning models can enable predictive analytics, allowing plantation managers to anticipate yield fluctuations, identify high-risk areas, and optimize workflows before losses occur.

1.2.3 Data-Driven Workflow Management

Workflow management in tea plantations involves coordinating plucking schedules, processing sequences, quality checks, and logistics for export. Studies in other industries show that visual programming environments (VPEs), domain-specific languages (DSLs), and drag-and-drop interfaces can empower non-technical domain experts to manage complex processes efficiently [14], [15]. Applying these concepts to tea plantations allows field supervisors, factory managers, and logistics coordinators to define, monitor, and optimize workflows without needing advanced programming knowledge. Furthermore, integrating predictive models into these systems enables dynamic adjustments to workflows in response to real-time data, such as sudden climate changes, labor shortages, or market demand fluctuations.

1.2.4 Predictive Modeling and Machine Learning

Machine learning models have been applied globally for crop yield prediction, pest forecasting, and quality assessment [16]. Techniques such as Random Forest, Support Vector Machines, and Neural Networks analyze large datasets from sensors, satellites, and historical production records. Studies indicate that predictive analytics can significantly reduce leaf damage, improve quality grading, and optimize harvesting schedules. While extensively used in countries like China and India, adaptation to Sri Lankan tea plantations requires calibration for local conditions, including elevation-based variability, microclimates, and labor patterns. Incorporating predictive modeling into decision-support systems can bridge the gap between traditional empirical methods and modern precision agriculture.

1.2.5 Global Comparisons and Best Practices

Globally, leading tea producers such as China, India, and Kenya have successfully integrated digital monitoring, mechanization, and data analytics to enhance productivity and maintain competitive advantage [17], [18]. For example, precision agriculture in China uses remote sensing and IoT to monitor leaf health and predict pest outbreaks, while India's high-tech plantations employ automated plucking and AI-driven quality assessment systems. Comparatively, Sri Lanka's plantations, particularly high-grown regions, face challenges in scaling such solutions due to terrain, labor limitations, and fragmented workflow management. Studying these global best practices highlights opportunities for Sri Lanka to adopt smart farming technologies in a context-sensitive manner, combining traditional knowledge with advanced data-driven methods.

Research Gap

Despite the significant economic and cultural importance of tea in Sri Lanka, the plucking process remains largely dependent on manual labor, with quality assessment relying on human expertise. Existing studies on tea grading and classification primarily focus on post-harvest evaluation, such as sorting, chemical analysis, and leaf grading after processing. While these approaches help maintain final product quality, they overlook the initial stage of leaf plucking, where the quality of harvest is first determined. Very few studies address the classification of tea leaves during plucking into categories such as under-plucked, correctly plucked, and over-plucked, despite its direct influence on both yield and final product quality. Moreover, conventional quality checks often lack consistency and are prone to human error, highlighting the need for automated systems that can provide reliable, real-time feedback during the harvesting process.

Although image processing and machine learning have been applied successfully in agricultural automation for tasks like fruit ripeness detection, disease identification, and crop monitoring, there is a scarcity of research specifically tailored to tea plucking. Most existing studies rely on static datasets collected under controlled laboratory conditions, which do not accurately reflect the dynamic and variable environment of tea plantations. Factors such as changing lighting conditions, leaf orientation, background complexity, and worker movement present challenges for applying conventional methods directly to the field. Consequently, there is a clear need for robust, plantation-ready systems that can consistently capture high-quality images, classify plucking stages accurately, and adapt to real-world conditions.

Furthermore, the integration of plucking stage classification with workforce management remains largely unexplored. Current research rarely links machine learning outputs to individual worker performance or efficiency metrics, which limits the practical application of automated classification in improving operational management. A comprehensive system that combines a conveyor-based image capture setup, machine learning-driven classification, and real-time tracking of worker productivity would not only enhance plucking accuracy and leaf quality but also support performance evaluation, workforce optimization, and continuous process improvement. Addressing this gap will enable the development of end-to-end solutions that bridge technological innovation with practical plantation management, ultimately benefiting both tea producers and the broader industry by improving quality, efficiency, and sustainability.

Research Problem

Tea production remains one of the most vital industries in Sri Lanka, contributing significantly to the country's economy and global reputation. A critical factor influencing both quality and yield in tea production is the plucking process, which traditionally relies on manual labor. However, this process is highly subjective, as workers' plucking accuracy varies widely depending on their skill level, fatigue, and attention. Incorrect plucking—whether under-plucking, over-plucking, or inconsistent plucking—directly impacts leaf quality, overall yield, and ultimately market competitiveness. Despite the significance of this stage, most tea estates continue to depend on conventional supervision methods, which are labor-intensive, prone to human bias, and incapable of providing real-time, data-driven feedback. This results in missed opportunities for ensuring consistency, maintaining premium quality standards, and improving workforce efficiency.

Recent developments in machine learning, computer vision, and image processing have opened up possibilities for automating agricultural quality assessment. In global contexts, systems have been developed for fruit grading, crop disease detection, and yield prediction, demonstrating significant success. However, existing research and solutions in the tea domain remain limited, fragmented, and often confined to experimental laboratory settings. For plucking stage detection in particular, most attempts rely on manual image assessment or broad leaf-quality grading without addressing the nuances of plucking correctness. Moreover, few systems incorporate worker identification alongside quality detection, meaning that the linkage between individual performance and quality outcomes is either absent or tracked through inefficient manual methods. This gap limits the ability of estate managers to make data-driven decisions about labor efficiency and targeted training.

Additionally, existing image-processing solutions in agriculture are often not optimized for real-time deployment in field conditions. Issues such as inconsistent lighting, leaf overlap, and motion on conveyor belts make classification tasks more challenging. Many current approaches also fail to integrate seamlessly into broader plantation management workflows, leaving a gap between detection and actionable workforce insights. In particular, the absence of systems that combine accurate classification of plucking stages, worker-level identification, and long-term performance analytics means that the tea industry continues to miss out on opportunities for digital transformation. As global competition in the tea market intensifies, Sri Lanka must adopt such innovations to preserve its high standards while improving productivity and workforce management.

This research addresses these gaps by developing an automated system for detecting and classifying tea plucking stages using machine learning and image processing, integrated with worker identification and performance tracking. By combining a conveyor-based image capture setup with advanced classification algorithms, the system ensures consistency and accuracy in plucking evaluation. Worker-level linkage enables states to monitor individual contributions, while aggregated efficiency analytics provide actionable insights for workforce

optimization. Unlike existing fragmented solutions, this integrated approach bridges the divide between quality assurance and labor management, paving the way for practical deployment and sustainable industry modernization.

Objectives

2.1 Main Objective

The primary objective of this research is to design and develop an automated system that can accurately detect and classify tea plucking stages using machine learning and image processing techniques, while simultaneously integrating worker identification and performance tracking. Unlike traditional supervision methods, which rely on manual observation and are prone to human bias, the proposed system will deliver real-time, data-driven insights to ensure plucking quality, enhance workforce efficiency, and support informed decision-making in Sri Lankan tea plantations. Ultimately, this research aims to provide a scalable and practical solution that improves both product quality and operational sustainability in the tea industry.

2.2 Specific Objectives

- To design a robust image acquisition setup for consistent data collection.
A conveyor-based imaging system will be developed with controlled lighting and environmental conditions to capture high-resolution, standardized images of fresh tea leaves. This ensures reliability and minimizes variability caused by external plantation factors.
- To build a labeled dataset representing different plucking stages.
Images will be systematically collected and annotated to create a comprehensive dataset that covers under-plucked, correctly plucked, and over-plucked categories. This dataset will serve as the foundation for supervised learning and reliable model training.
- To develop an accurate and robust machine learning model for classification.
Advanced machine learning algorithms, supported by image processing techniques, will be implemented to classify tea plucking stages in real time. The model will be optimized to maintain high accuracy and robustness even under plantation conditions where variability in leaf presentation is common.
- To integrate worker identification for individual performance tracking.
Classification results will be linked with worker identities, allowing estate managers to monitor each Plucker's efficiency and accuracy. This integration provides actionable feedback that supports performance improvement and accountability.
- To analyze and visualize Plucker efficiency and workflow data.
The system will store, process, and present performance metrics in a structured manner. These analytics will assist plantation managers in workforce planning, identifying training needs, and ensuring continuous quality improvement.
- To validate the system for practical deployment in real plantation environments.

Field testing and evaluation will be conducted to ensure the solution is reliable, scalable, and seamlessly fits into existing tea harvesting workflows, enabling adoption at an industrial scale.

System Methodology

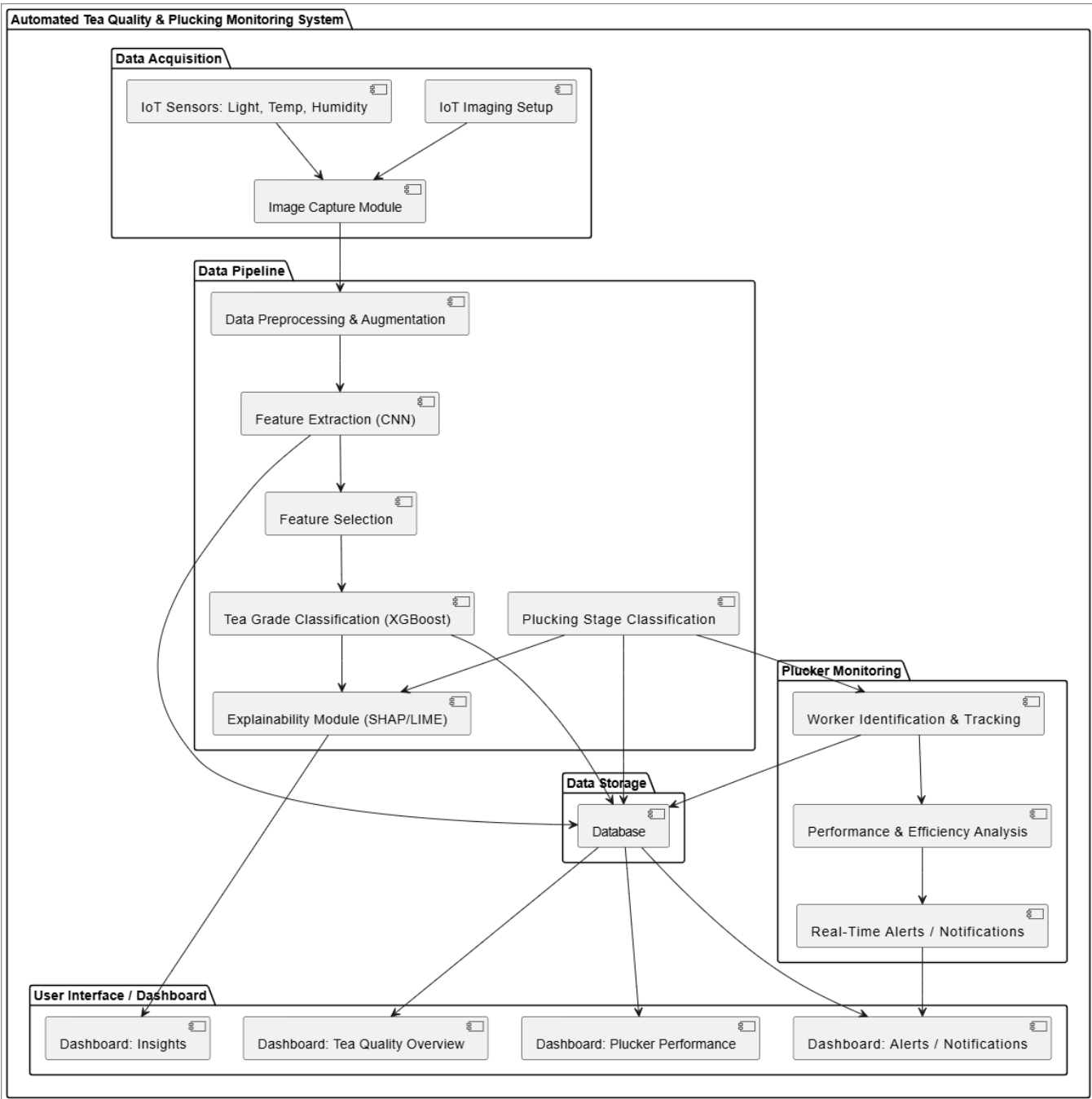


Figure 3 - System Overview

The strategy for this research is to integrate IoT technology, computer vision, and machine learning to automate the quality assessment of plucked tea leaves and track worker efficiency in Sri Lankan tea plantations. Traditional tea quality assessment relies on manual inspection, where workers or supervisors visually examine samples and classify the plucked leaves based on their condition. This process is labor-intensive, time-consuming, and subject to human error or subjective bias. Currently, operators evaluate a 100g sample and determine whether it meets export quality ($\geq 60\%$ good leaves), production quality (45–60%), or is unsuitable ($< 45\%$). The proposed system automates this evaluation using a conveyor belt for sample transport, an IoT-based camera for image capture, and machine learning models for classification and performance analysis.

The methodology establishes a step-by-step process from conceptual design to deployment, including iterative development, system optimization, data preprocessing, model training, integration with operational workflows, and field validation. The system will ensure high accuracy in leaf condition detection, allow real-time assessment, provide detailed feedback for worker performance, and support data-driven decision-making. By employing advanced image processing and machine learning methods, the system will handle variations in leaf orientation, lighting, occlusion, and debris while maintaining scalability for plantation-level deployment.

3.1 Application of Key Pillars within the Specialized Knowledge Domain

This study combines concepts from IoT, computer vision, machine learning, and operational analytics:

IoT and automation: The conveyor belt system with an integrated camera allows precise, repeatable sample handling. IoT-enabled devices ensure real-time capture and communication of image data to the processing system, reducing manual handling errors. Sensors monitor conveyor speed, sample placement, and camera alignment to guarantee consistent data acquisition.

Computer vision: High-resolution cameras capture images of tea leaves as they move along the conveyor. Image preprocessing techniques, such as background subtraction, color normalization, edge detection, and contour analysis, are applied to isolate leaves from twigs and debris. This enables accurate segmentation of leaf regions for feature extraction, ensuring that classification models can evaluate each leaf's condition reliably.

Machine learning: Deep learning architectures, particularly convolutional neural networks (CNNs) or hybrid models combining CNNs with ensemble classifiers, are employed to classify plucking stages. Models are trained to recognize good leaves, minor defects, tears, or debris, allowing for batch-level evaluation of quality percentages. These models account for environmental variability such as lighting changes or overlapping leaves, ensuring robust predictions.

Operational analytics and worker performance tracking: The system maps each tea Plucker's output by linking processed samples to individual workers via barcode, RFID, or manual

identification. Performance dashboards visualize metrics such as plucking efficiency, quality adherence, and consistency over time, enabling supervisors to provide targeted feedback, optimize workforce management, and reduce overall waste.

By combining these pillars, the system integrates automation, precision, and data-driven insights into the traditional tea harvesting workflow, enhancing operational efficiency while maintaining export-quality standards.

3.2 Application of Technologies in the Field of Application

The system uses current frameworks and hardware optimized for plantation conditions:

IoT-based sample handling: A conveyor belt system transports 100g samples of plucked tea leaves to the camera station. Load sensors ensure consistent sample weight and placement, while motor control adjusts speed to maintain image clarity.

Camera and image capture: A high-resolution industrial camera is installed above the conveyor belt. Controlled lighting ensures minimal shadows and consistent color representation. Images are captured in real time and transmitted to an edge device for preprocessing and classification.

Image preprocessing: OpenCV and other Python libraries handle leaf segmentation, removing twigs, stems, and debris. Techniques such as thresholding, morphological operations, and edge detection isolate leaf features for the classification model.

Classification model: A CNN or hybrid deep learning model is trained to identify leaf conditions. The model outputs the proportion of good leaves versus defective ones, classifying batches as export quality, production quality, or unsuitable. Pretrained models can be fine-tuned on plantation-specific data to increase accuracy.

Worker tracking and performance monitoring: Each batch processed is tagged to a specific worker using RFID cards, QR codes, or manual input. Real-time dashboards display metrics like average batch quality, number of batches processed per hour, and consistency scores.

Data storage and processing: Edge devices perform real-time analysis to reduce latency. Summary metrics and anonymized sample images are uploaded to a central server for archival, reporting, and further analytics.

These technologies are selected based on accuracy, scalability, and operational feasibility, enabling the system to function reliably in plantation environments without significant disruption to existing workflows.

3.3 How to Carry Out the Project

The research will adopt an iterative development life cycle:

- **Requirement analysis and design:**
Define technical requirements for conveyor speed, sample size, camera resolution, lighting conditions, and classification accuracy. Identify metrics for worker performance and dashboard design specifications.
- **Prototype development:**
Build the first conveyor and camera setup. Capture initial sample images to test preprocessing and classification feasibility. Validate whether images contain sufficient detail for accurate leaf assessment.
- **Data collection and preprocessing:**
Collect large datasets from multiple plucking sessions. Annotate images for good leaves, minor defects, and debris. Apply image augmentation to simulate varying lighting, angles, and occlusions, enhancing model robustness.
- **Model training and optimization:**
Train CNN-based or hybrid models on labeled datasets. Perform hyperparameter tuning, cross-validation, and pruning to achieve high accuracy with low inference latency. Optimize for edge devices to ensure real-time processing.
- **Integration with operational workflow:**
Connect conveyor, camera, and edge device into a seamless pipeline. Tag batches with worker identifiers. Ensure the system outputs quality percentages and flags batches that fail to meet export or production standards.
- **Testing and tuning:**
Conduct controlled tests to evaluate batch classification accuracy under varying lighting, leaf density, and debris conditions. Adjust conveyor speed, lighting, and camera positioning to maximize detection reliability.
- **Deployment and monitoring:**
Install the system in plantation facilities. Train supervisors to use dashboards for real-time monitoring of worker performance and batch quality. Collect feedback for further iterative improvements.

3.4 Tasks and Sub-Tasks Identified to Meet Objectives

1. Conveyor and camera system development:
 - Design sample transport system for consistent 100g leaf batches.
 - Integrate IoT sensors to monitor weight, speed, and alignment.
 - Test stability and durability for continuous plantation use.

2. Image capture and preprocessing:
 - Implement high-resolution camera with controlled lighting.
 - Apply image preprocessing: segmentation, thresholding, noise removal.
 - Ensure reproducibility of leaf detection across batches.

3. Leaf quality classification:
 - Select machine learning model architecture (CNN, hybrid CNN + ensemble).
 - Train models on labeled datasets for good, production-quality, and defective batches.
 - Optimize inference speed for real-time classification on edge devices.

4. Worker tracking and analytics:
 - Implement identification tagging for workers (RFID, QR, manual).
 - Build dashboards showing quality metrics, efficiency, and trends.
 - Enable export of reports for managerial decision-making.

5. System integration and field validation:

- Connect conveyor, camera, edge device, and dashboard into a single pipeline.
- Conduct field trials in real plantation conditions.
- Monitor batch classification accuracy, worker efficiency, and system reliability.

3.5 Data Requirements and Collection Methods

The system requires a large, heterogeneous image dataset representing real-world plucked tea samples:

- Images should cover all plucking stages: good leaves, minor tears, excessive debris, and twigs.
- Data should include variability in lighting, leaf orientation, overlapping leaves, and background noise.
- Collection involves capturing batches from multiple plantations, seasons, and weather conditions.
- Data preprocessing includes segmentation, normalization, resizing, and augmentation (rotation, brightness adjustment, flipping) to improve model robustness.
- Anonymized images are stored to maintain privacy of plantation-specific operational data.

Publicly available datasets may be used for initial model training, but fine-tuning on plantation-specific images ensures higher accuracy in real deployment conditions.

3.6 Project Requirements

3.6.1 Functional Requirements

- Batch quality classification: The system must accurately classify 100g leaf samples as export, production, or defective in real-time.
- Integration with worker tracking: Each batch's result must be linked to a specific worker for performance evaluation.
- Real-time analysis: Batch classification must be completed in seconds to support continuous workflow.
- Multiple batch handling: The system must simultaneously evaluate multiple samples without performance degradation.

3.6.2 User Requirements

- Simple dashboard interface for supervisors to monitor quality and efficiency.
- Minimal configuration: system should work with standard plantation infrastructure.
- Transparency: display classification results, confidence levels, and potential errors.

3.6.3 System Requirements

- Optimized for mid-range edge computing devices.
- Compatibility with standard camera formats.
- Scalable architecture for multiple conveyor-camera setups.

3.6.4 Non-Functional Requirements

- Accuracy and robustness: High classification accuracy across leaf conditions and lighting scenarios.
- Reliability: Continuous operation with failover support.
- Flexibility: Easy integration of updated models or new performance metrics.

3.6.5 Use Case Diagram

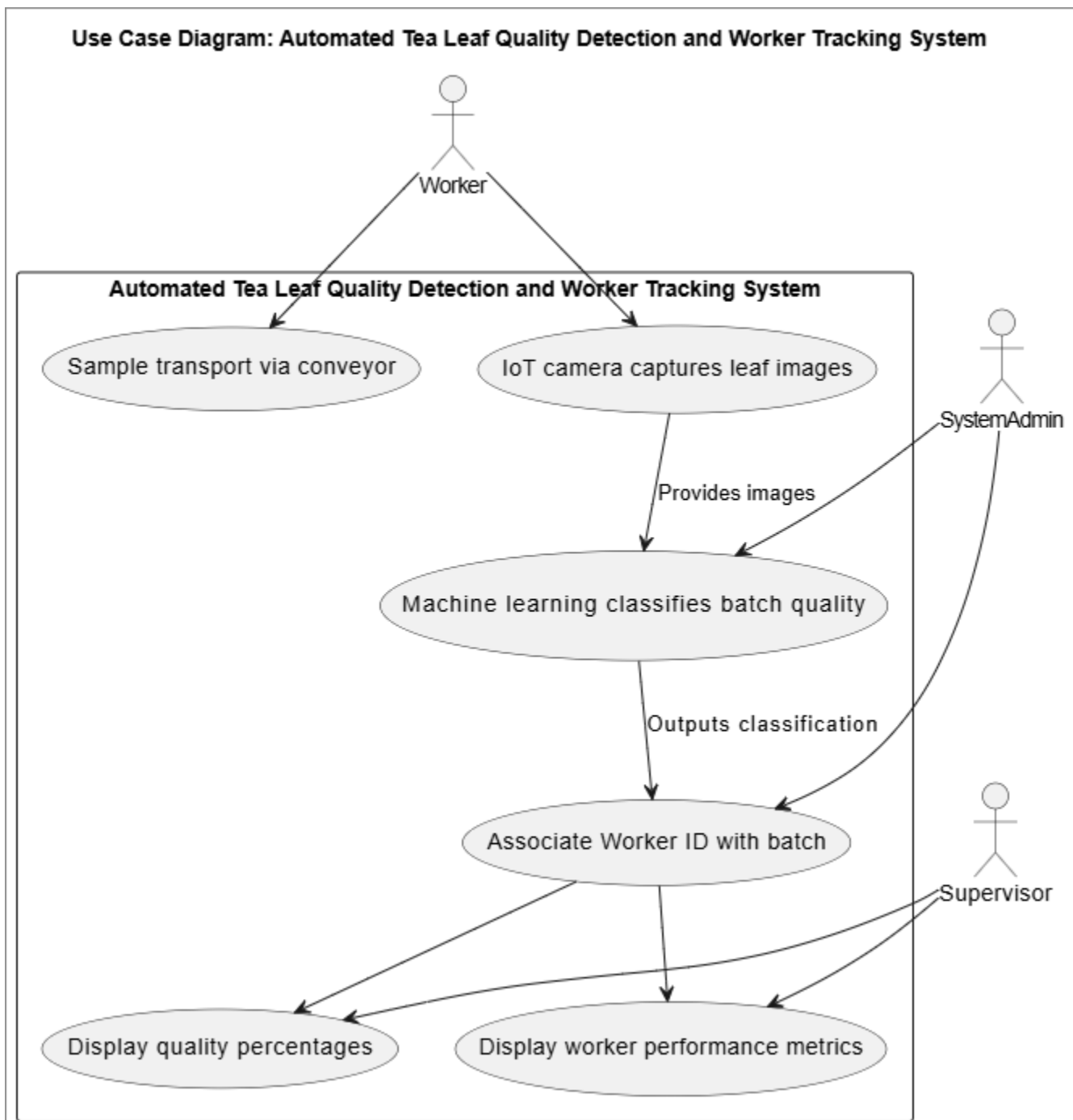


Figure 4 - Use Case Diagram

Test Cases (tentative)

Test Area	Test Case ID	Description	Input Data	Expected Outcome
Camera and Conveyor System	TC-I.I	Verify camera captures batch correctly	Batch placed on conveyor	High-quality image captured
	TC-1.2	Check lighting adjustment	Dim/bright environment	System corrects or warns
	TC-1.3	Test conveyor alignment	Misaligned batch	System repositions or alerts
	TC- 1.4	Validate batch weight monitoring	Sample <100g or > 100g	System flags weight inconsistency
Leaf Classification	TC-2.1	Verify correct batch classification	Sample images	Batch labeled export/production/defective
	TC-2.2	Test detection under occlusion	Overlapping leaves	Model correctly identifies leaf conditions
	TC-2.3	Validate debris detection	Twigs/debris present	Debris correctly excluded
	TC-2.4	Check speed of classification	Continuous batches	Real-time classification (<2s per batch)

3.7 Anticipated Conclusion

By following this methodology, the project will deliver an automated, high-precision tea leaf quality assessment system capable of real-time batch classification and worker performance tracking. Expected outcomes include:

- Improved consistency and accuracy in leaf quality evaluation.
- Reduced reliance on manual inspection, saving time and labor costs.
- Enhanced decision-making for production and export quality selection.
- Data-driven insights on worker performance for targeted training and incentives.
- Scalable solution deployable across multiple plantations with minimal adaptation.

The system bridges IoT, computer vision, and machine learning with operational analytics, offering a sustainable, efficient, and technologically advanced solution to Sri Lanka's tea harvesting and processing workflow.

COMMERCIALIZATION ASPECTS OF THE PRODUCT

The proposed AI and Computer Vision-Based Automated Tea Leaf Quality Evaluation System is designed to serve tea estates, processing factories, and exporters, providing a highly efficient tool for quality assurance and operational management. By integrating IoT-based imaging, conveyor-based sampling, and machine learning classification, the system automates the traditionally labor-intensive process of assessing plucked tea leaves, reducing human error and improving throughput. Its commercial value lies in enabling tea producers to consistently classify batches based on leaf condition, optimize workforce allocation, minimize production delays, and ensure compliance with export-grade standards. The solution is adaptable to diverse scales of tea production, from small plantations to large commercial factories, offering significant opportunities for sustainable commercialization.

5.1. Market Space

Tea Processing Market

The global tea industry continues to grow steadily, with emphasis on quality and consistency becoming increasingly critical for both domestic markets and international export. Automation of quality assessment is emerging as a strategic requirement, as producers seek scalable, accurate, and cost-effective alternatives to manual inspection. AI-driven systems that can evaluate batch quality in real time, track worker contributions, and provide actionable insights are becoming highly desirable. This system aligns with current trends, providing a timely solution to address industry inefficiencies.

Market Size

The market for automated tea processing and quality control spans several billion dollars globally. In key tea-producing regions, including Sri Lanka, India, and China, most factories rely heavily on manual sorting, which is prone to variability, fatigue, and subjective judgment. Adoption of automated, AI-based systems in these regions is still in its early stages, representing an ideal entry point for a cost-effective and technologically advanced solution. By providing measurable improvements in accuracy, consistency, and productivity, this system positions itself as a valuable investment for both small-scale and large-scale producers aiming to maintain competitive quality standards.

Existing Systems

Current quality control methods rely primarily on human visual inspection and, in some cases, static imaging or basic sorting machinery. These approaches are limited in their ability to detect subtle variations in leaf quality, including the presence of tears, twigs, or other debris. Additionally, they offer no means of integrating worker performance tracking, and operational efficiency is constrained by manual labor. The proposed automated system fills this gap by providing precise, real-time batch evaluation, tracking worker activity, and delivering actionable metrics for management to optimize production.

Competition

The competitive landscape includes manufacturers of traditional sorting machinery and emerging AI-driven agricultural technology providers. While some offer basic automated sorting solutions, few integrate full automation with IoT imaging, machine learning classification, conveyor-based sampling, and worker tracking. This system differentiates itself by combining these features into a cohesive, scalable solution capable of accurate batch classification, consistent quality assurance, and improved operational visibility. Its competitive advantage arises from reducing labor dependency, increasing productivity, and providing actionable insights without significant capital investment in specialized infrastructure.

5.2. Revenue Model and Commercial Strategy

Licensing Model – Tea estates and processing factories can acquire licenses to deploy the system across multiple production lines or facilities. Licensing offers a predictable revenue stream and scales efficiently with the expansion of client operations.

Subscription Model (Software-as-a-Service) – Offering the platform on a subscription basis makes the solution accessible to smaller or budget-conscious producers. Monthly or annual plans can include software updates, analytics dashboards, and support services, ensuring that clients maintain access to the latest AI models and system enhancements.

Hardware + Software Bundling – For locations with limited infrastructure, the system can be offered as a turnkey solution bundled with conveyor units, IoT cameras, and edge-computing devices. This approach provides an end-to-end solution for automated tea leaf evaluation while opening markets among smaller or modernizing estates.

Data-Driven Insights as a Service – Aggregated and anonymized data on batch quality and worker performance can be sold as analytics to industry stakeholders, such as tea cooperatives, consultants, or exporters. Insights derived from AI predictions can inform decisions regarding harvesting schedules, processing techniques, and export readiness while maintaining data privacy and protecting operational details.

Consulting and Integration Services – In addition to the core system, consulting services for factory integration, staff training, and workflow optimization can be offered. This adds an additional revenue stream and enhances the perceived value of the system by ensuring clients maximize operational benefits from automation.

Continuous Improvement and Model Updates – Clients can be offered subscription-based updates for AI models and software optimization, allowing them to maintain high-quality standards over time. Continuous learning and adaptation ensure that the system remains effective under evolving processing conditions, such as variations in leaf types, environmental factors, and seasonal differences in harvest quality.

This multi-pronged commercialization strategy ensures that the system can be offered to a wide range of customers, from small plantation owners to large-scale commercial factories, while maintaining affordability, scalability, and operational relevance. By addressing the core challenges of manual quality inspection—labor intensity, inconsistency, and limited traceability—this solution provides both immediate efficiency gains and long-term strategic value.

Sustainability and Market Adoption – The system contributes to sustainable production by reducing waste and improving export-quality yield, strengthening the reputation of adopting estates and factories in competitive markets. Early adopters can position themselves as technologically advanced, quality-focused producers, enhancing both domestic and international competitiveness.

Expansion Potential – Beyond tea, the system can be adapted for other agricultural products where quality inspection is critical, including coffee, herbs, and specialty leaves. This versatility increases long-term commercialization opportunities, ensuring that the underlying AI and IoT technology remains relevant across multiple segments of the agricultural industry.

Competitive Advantage – By combining automated quality evaluation with worker tracking, the product delivers operational transparency and accountability, reducing errors and optimizing workforce performance. Its modular and scalable design allows adoption across facilities of varying sizes without significant capital expenditure, creating a competitive advantage in both cost-effectiveness and technological sophistication.

Overall, the AI and Computer Vision-Based Automated Tea Leaf Quality Evaluation System offers a transformative approach to tea processing, combining operational efficiency, precise quality control, worker accountability, and data-driven insights. Its commercial strategy positions it to be a sustainable, high-value product for tea producers seeking automation, standardization, and enhanced productivity while ensuring consistent quality for domestic and export markets.

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